JIN
Teeth Surface Forming Machine

Operation Manual
(Electric Diagram)

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1. About automatic driving

1-1. Automatic startup procedure

Start procedure of automatic driving is as follows.

1) Turn ON this equipment power switch.

2) Reset the "EMERGENCY STOP" button on the operation panel and back side of machine (turn the switch clockwise).

3) Push the "MASTER" illuminated push button switch, and confirm lighting of the lamp. When driving preparations do not enter, confirm next.
   - Is PLC and Safety PLC in a panel running?
   - Does not the light curtain blocked?
   - Does not abnormality of servo driver occur?
   - Does not abnormality of Rotation Refresh Converter occur?

4) Confirm whether abnormality does not occur.
   When abnormality occurs, a red light of facility signal tower turns on.
   Confirm contents with a " Fault Description " screen of touch panel, and push the "RESET" button after cancellation of the cause, and remove abnormality.

5) Make the "CYCLE" select switch "AUTO" side.

6) Please select "TYPE" with TYPE SELECT screen of touch panel.
   Please push a "type number change" switch if different from type number to hope for. A type number selection window is displayed. Enter the desired model number and press the "Change model number" switch for 2 seconds to change the model number.

7) Set a work, and please insert a lower shaft serration.

8) Confirm comment of " Can start automatic operation." with "MAIN" screen of touch panel. When comment except this appears, please obey instructions of the comment.

9) Automatic driving is started when knocked on “lever switch”.
   If work is completed, and all actuators return to an original position, it is stopped.

1-2. Automatic driving stop method

Stop method of automatic driving has the following method.

1) Switch off this equipment.
   An arrangement stops instantly at the place. There are some cylinders using a solenoid valve of two positions, and warn those actuators because those cannot stop in mid career. Do not use this method as much as possible.

2) Push the " EMERGENCY STOP " button.
   Preparing it is running out of driving, and equipment stops instantly at the place.
   There is danger to a worker or a work, and use this method when it is urgent.

3) Light curtain block.
Preparing it is running out of driving, and equipment stops instantly at the place.

4) Make the "CYCLE" select switch "MANU." side.
Automatic driving stops immediately.

5) Pushing the “HOME POSITION” push button switch.
Automatic driving stops immediately. But some cylinders using a solenoid valve of two positions cannot stop in mid career.

2. About return driving

You can return an actuator to machine original position by carrying out return driving.

2-1. A return startup procedure

The start procedure of return driving is as follows.

1) Turn ON this equipment power switch.

2) Reset the "EMERGENCY STOP " button on the operation panel and back side of machine (turn the switch clockwise).

3) Push the "MASTER" illuminated push button switch, and confirm lighting of the lamp. When driving preparations do not enter, confirm next.
- Is PLC in a panel running?
- Does not the light curtain blocked?
- Does not abnormality of servo driver occur?
- Does not abnormality of Rotation Refresh Converter occur?
- Does not abnormality of L/S torque?

4) Confirm whether abnormality does not occur.
When abnormality occurs, a red light of facility signal tower turns on.
Confirm contents with a "Fault Description" screen of touch panel, and push the "RESET " button after cancellation of the cause, and remove abnormality.

5) Confirm whether NG does not occur.
When NG occurs, a red light of facility signal tower turns on.
Perform NG measures.

6) Make the "CYCLE" select switch "MANUAL" side.

7) Confirm comment expression of "Can start Return operation" in "main screen "of touch panel. When comment except this appears, obey instructions of the comment.

8) Push the "HOME POSITION" push button switch, then auto cycle start.
When the equipment becomes original position, it stops.
If the equipment does not become original position for 30 seconds, it gives up and stops.

2-2. Return driving stop method

As for return driving, usually when the equipment becomes original position, it stops automatically.
The method to stop return driving on the way has the following method.

1) Switch off this equipment.
   An arrangement stops instantly at the place. Do not use this method as much as possible.

2) Push the "EMERGENCY STOP" button.
   Preparing it is running out of driving, and equipment stops instantly at the place.
   There is danger to a worker or a work, and use this method when it is urgent.

3) Light curtain block.
   Preparing it is running out of driving, and equipment stops instantly at the place.

4) Make the "CYCLE" select switch "MANU." side.
   Return driving stops immediately.

3. About manual operation

3-1. The startup procedure of manual operation

Startup procedure of manual operation is as follows.

1) Turn ON this equipment power switch.

2) Reset the " EMERGENCY STOP" button on the operation panel and back side of machine (turn the switch clockwise).

3) Push the "MASTER" illuminated push button switch, and confirm lighting of the lamp.
When driving preparations do not enter, confirm next.
- Is PLC in a panel running?
- Does not the light curtain blocked?
- Does not servo amp abnormality occur?
- Does not abnormality of Rotation Refresh Converter occur?

4) Make the "CYCLE" select switch "MANU" side.

5) Choose an item (an actuator), which you want to operate with a screen of "Manual Operation" and press the "EXECUTE" illuminated button to operate the actuator.
When movement condition of the chosen actuator is concluded, lamps of these illuminated buttons flash on and off, and the lamp of corresponding lights it when it arrives at a movement edge.
＊Manual operation does not start it when blocked light curtain.
In addition, it does not work unless it satisfies movement condition.
An actuator stops immediately when get off these illuminated push button swatches.

6) An actuator that inching is possible stops immediately when get off these illuminated push button swatches.

7) An actuator block with a light curtain during an operation. During automatic driving equipment stops immediately when they push this switch.

4. Indication light / operation switch explanation

Functions of indicators and operation switches on the control box and the control panel are as follows.

1) The “POWER” indication light

It is with a main circuit breaker, and a control power supply turns on during electricity.

2) The "CYCLE" select switch

It is as follows in a choice state.

"MANUAL”: manual operation each one operation is enabled.

"AUTO": automatic driving is enabled.

3) The " MASTER " illuminated push button switch

Switch:

Turn on driving preparations.

Power is supplied, and / return / automatic driving worked by hand is enabled.

Condition:

The "EMERGENCY STOP" button of a control panel and back side art not pushed into.

・Is PLC in a panel running?

・Does not the light curtain blocked?

・Does not servo amp abnormality occur?

・Does not abnormality of Rotation Refresh Converter occur?

Lamp:

Turn on among driving preparations ON.

4) The "EMERGENCY STOP" lock type push button switch

It is running out of driving and stops all power supply. Establish it in a control panel and the vicinity of a back safety door.

5) The "RESET" push button switch

When an abnormal cause is removed, abnormality is reset when you push this switch. The warning buzzer stops when you push this switch when an abnormal cause is not removed.

6) The "HOME POSITION" type push button switch

When “CYCLE” is selected “MANU”, you push this switch. And return driving starts. During automatic driving equipment stops immediately when they push this switch.

7) The "EXECUTE" illuminated push button switch

Switch:

Operate an operation item (an actuator) chosen with a screen of "Manual Operation" of touch panel to the original position side.

Condition:

Driving preparations are ON, and "MANU." is chosen with the "CYCLE" selecting switch.

There are conditions for each operation item.

Lamp:

When the original position side operation condition of a chosen operation item is concluded, it flashes on and off.

It turns on at time of the chosen operation item is the original position end.

In addition, turn on when you push a "Lamp Check" switch of touch panel "Setting Menu Screen".

8) Buzzer

When a Lever switch was able to leave ON in the state that in heterology or automatic driving cannot carry out, it is sounded.

In addition, sound when auto driving passed with possible condition more than five seconds.

A buzzer stops when pushes a "reset" switch.

9) “ACCESS OK” Lamp

When they do not work, an actuator and a servomotor turn on Midori. I turn on red during the movement of facilities. The driving preparations do not fail in a rye toe curtain at the time of the green lighting even if they shade the light, but driving preparations fail in light car ten at the time of the red lighting when they shade the light.

5. About the touch panel screen

There is a screen such as follows in touch panel.

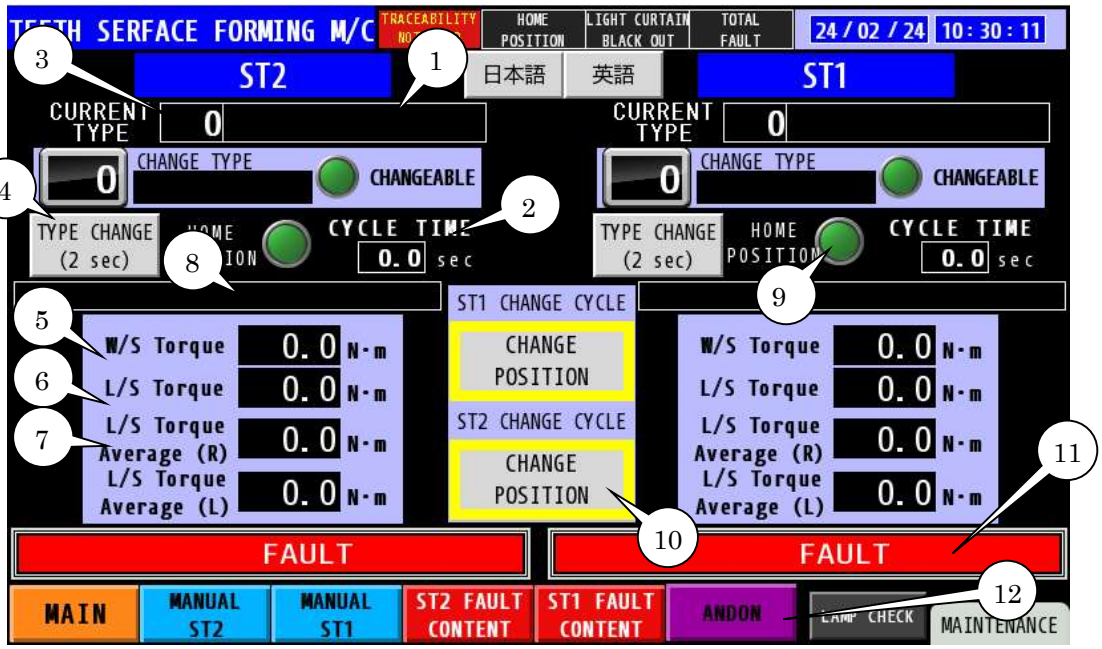
Screens not requiring password

- MAIN SCREEN
- MANUAL OPEARTION SCREEN
- PLC DEVICE MONITOR SCREEN
- COUNTER SCREEN
- SPECIFICATIONS MONITOR SCREEN
- ABNORMAL DESCRIPTION SCREEN
- ABNORMAL HISTORY SCREEN

Screens that cannot be displayed unless password is input

- COMMON SPECIFICATIONS SETUP SCREEN
- SPECIFICATIONS SETUP SCREEN
- COPY OF SPECIFICATIONS SCREEN
- TYPE NAME REGISTRATION SCREEN
- L/S TORQUE CALIBRATION SCREEN

5-1. Main screen



1) It displays the name selected for product currently.

2) It displays cycle time of the machine.

3) It displays the work type number
for product currently.

4) Type change switch
Please press the model number you want to select.
Press for 2 seconds to change the model number.

5) Display current input shaft torque.
Display current input shaft torque.

6) Display current load shaft torque.
Load shaft torque is output at +-10V by AS2603. Become 200N / m at 10V.

7) Display Average L/S Torque
Revise load torque during forming. Display average load torque between revision.

8) DMC code display.

9) System home position lump
The station turns on at a home position.

10) Tool change cycle manual select switch and lamp
This button lights, and when machine home position, the tool change cycle can be started.

11) Comment indication
Here, it displays various comments to help operation of an operator.
(Refer to 5-1-1."Main" screen comment list)

12) Touch panel screen changing switch

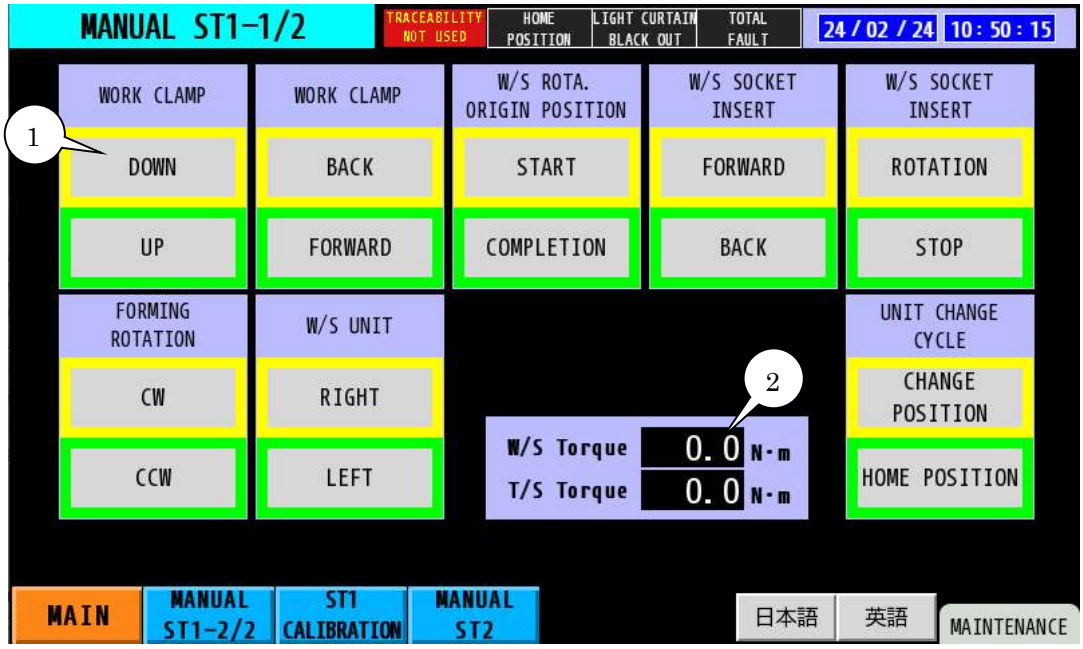
5-1-1. Main screen comment list

Comments on the Main screen are allocated to the PLC internal relays ST1 M640 to M671, ST2 M1640 to M1671. When an internal relay turns ON, the comment corresponding to the internal relay is displayed.
If multiple internal relays should turn ON simultaneously, switch comment at an interval for three seconds.

Main screen comment list

ST1 Device	ST2 Device	Comment
M640	M1640	Fault Occurred.
M641	M1641	Please press "MASTER ON" button.
M642	M1642	Emergency stop button is pressed.
M643	M1643	Manual operation.
M644	M1644	Light curtain is being blocked.
M645	M1645	During safety maintenance
M646	M1646	Vacuum filter check timing
M647	M1647	L/S Torque NG
M648	M1648	Please check the air dryer power.
M649	M1649	W/S Torque NG
M650	M1650	Automatic off NG
M651	M1651	Please change W/S socket.
M652	M1652	Please change L/S socket.
M653	M1653	Lack of Tonner forecast
M654	M1654	Please set a work-piece.
M655	M1655	Please take out a marker pen.
M656	M1656	Please take out a work-piece.
M657	M1657	Please insert the Serration.
M658	M1658	Please return to home position.
M659	M1659	Can start automatic operation.
M660	M1660	Please start warm-up cycle
M661	M1661	Turn Setting switch off.
M662	M1662	During warm-up cycle
M663	M1663	During safety maintenance
M664	M1664	During auto run cycle
M665	M1665	During return cycle
M666	M1666	
M667	M1667	Read Data Matrix code
M668	M1668	TARCEABILITY-NOT OK TO RUN!
M669	M1669	TYPE MISMATCH WITH SERVER
M670	M1670	Waiting response from server
M671	M1671	PLC Battery Error

5-2. Manual Operation

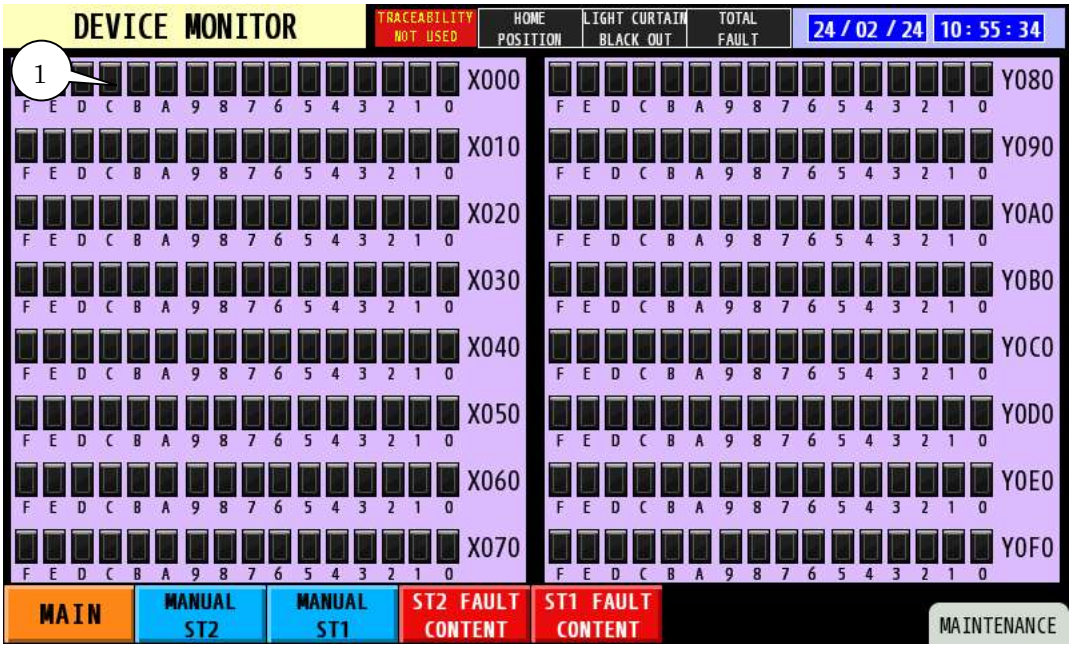


1) Combined illuminated type selector PB switch + collective indicator lamp
There are switches to select items to be selected (actuators) and lamps to indicate selection of the items.
When the run preparation is ON and the "MANUAL" mode is selected on the "CYCLE" selector switch, push an item to be operated. An operation items is selected, and the lamp for the item comes on to indicate the item is selected.
When an operation items is selected, individual operation becomes available by pressing the "EXECUTE" illuminated type pushbutton switch or "RETURN" illuminated type pushbutton switch. However, refer to above-mentioned "3-2. Manual operation condition lists" because operating conditions are specified per operation item.
In addition, indicator lamps are provided on the operation side and the original position side ends of each item, so that the lamp comes on when the applicable sensor turns ON.

2) Load Torque Display
Display current load torque value.

5-3. PLC Device Monitor

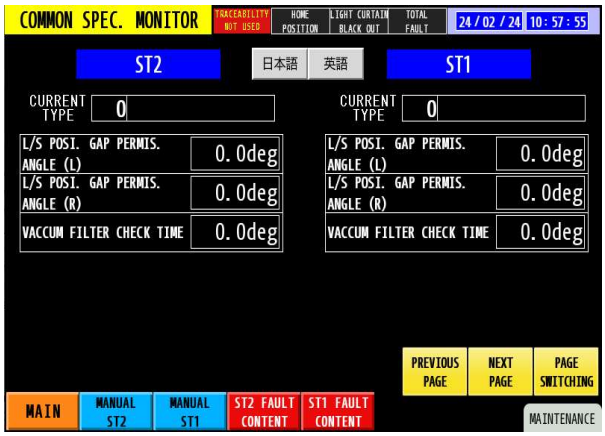
Almost all devices possessed by the PLC can be monitored on this screen. Monitoring is performed at the unit of 1 word (16 bits). Two different kinds of devices are monitored the ON/OFF status, hexadecimal numbers and decimal numbers of respective bits at the same time.

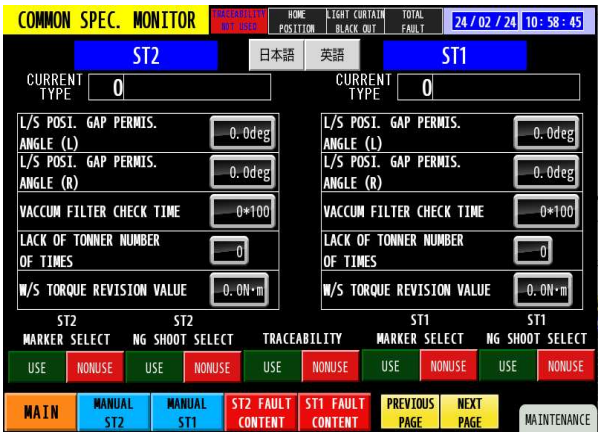


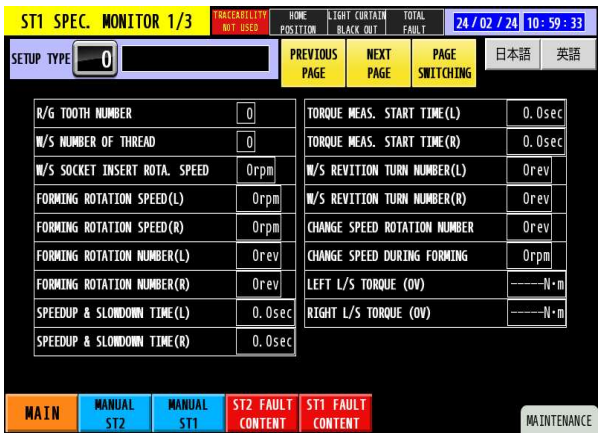
1) A lamp displays an ON/OFF state of a bit unit. (ON = green)

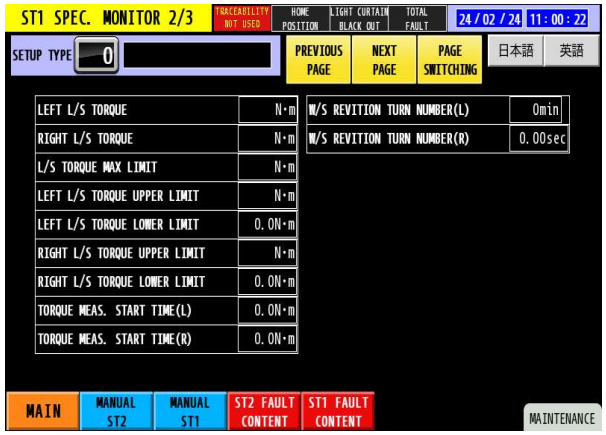
5-4. Common Specifications Setup Screen

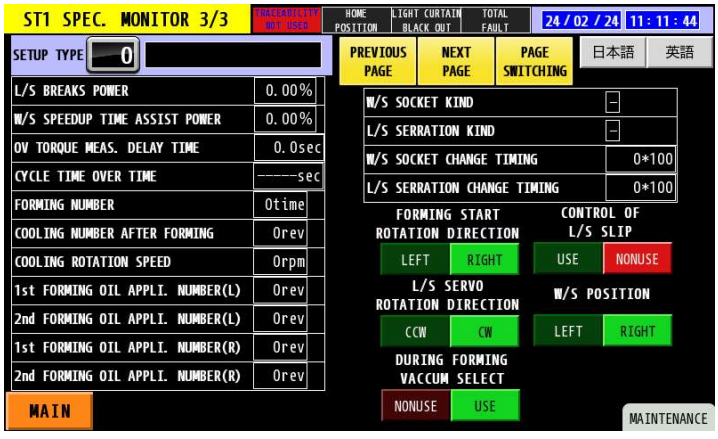
Can refer to various set value of later description 5-8. Common Specifications Setup screen with this screen. But can't change set value in this screen.
For shifting to the setting screen (Specifications Setup), a password is required, but no input of a password is required to this screen.
For contents of each item, refer to Setup screen.





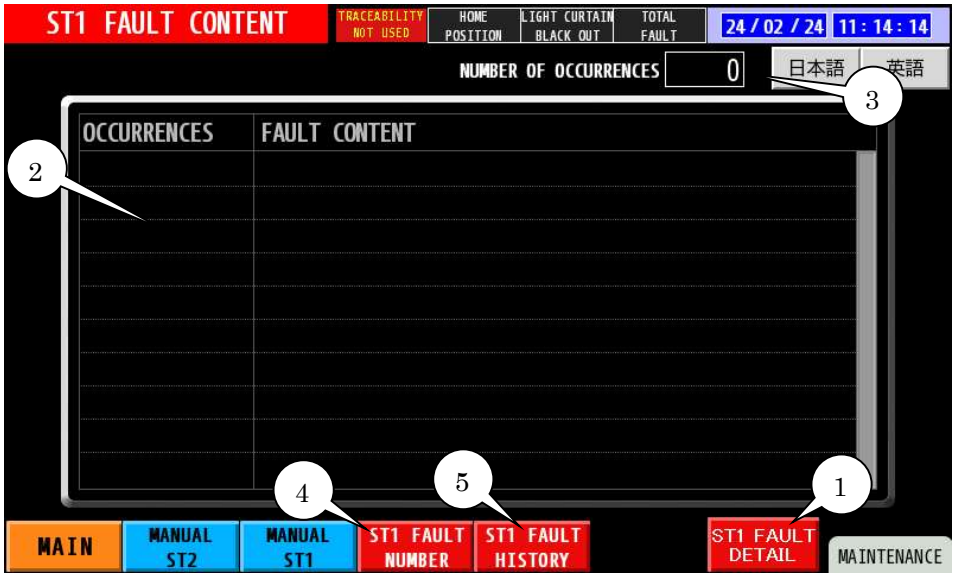






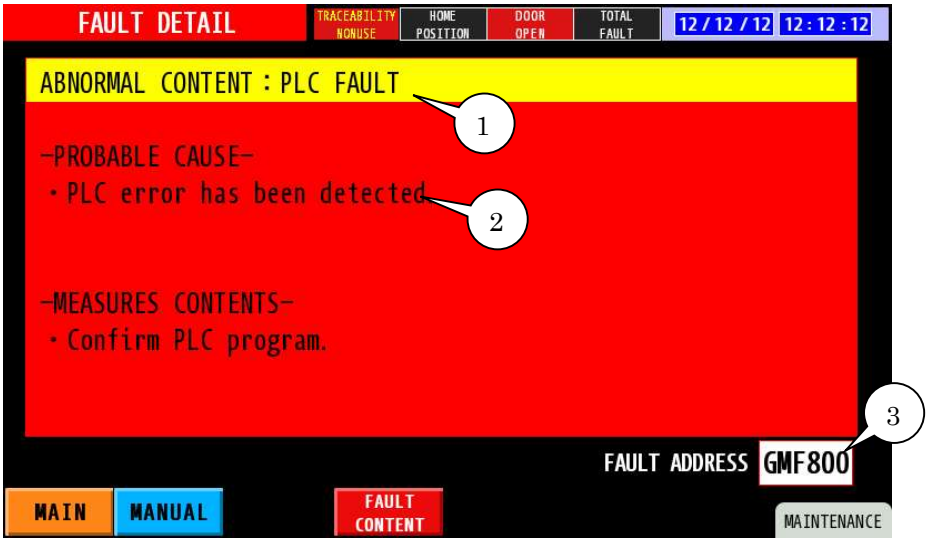
5-5. Fault Description

5-5-1. Fault Content



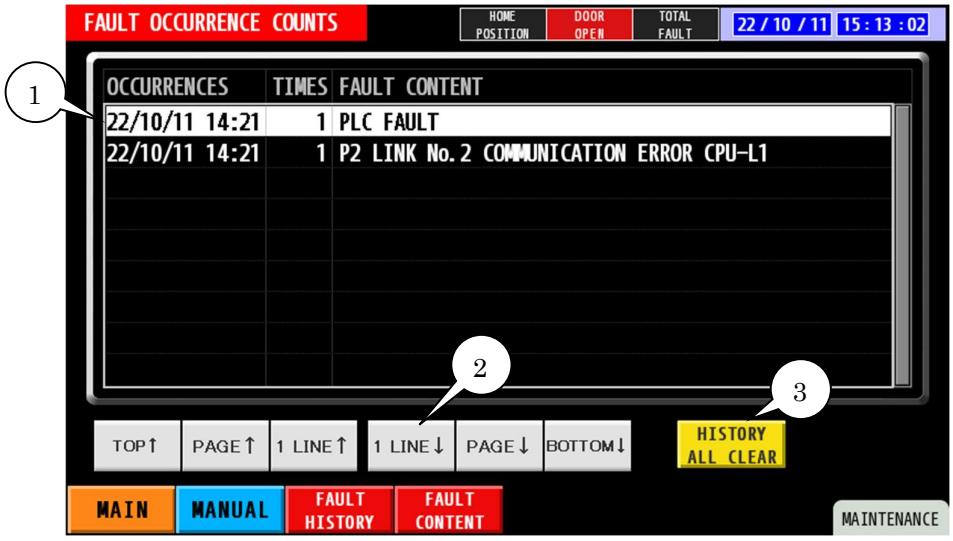
1) Displays the total number of abnormalities that are currently occurring.
2) The fault contents that are currently occurring are displayed
3) Displays the contents and details of the selected anomaly in the currently occurring anomaly.
4) Touch this switch to move to the FAULT NUMBER screen.
5) Touch this switch to move to the FAULT HISTORY screen.

5-5-2. Fault Detail



1) The abnormal contents that are currently occurring are displayed.
2) Displays the contents and details of the selected anomaly in the currently occurring anomaly.
3) The PLC internal relay number corresponding to the abnormality displayed in 2).

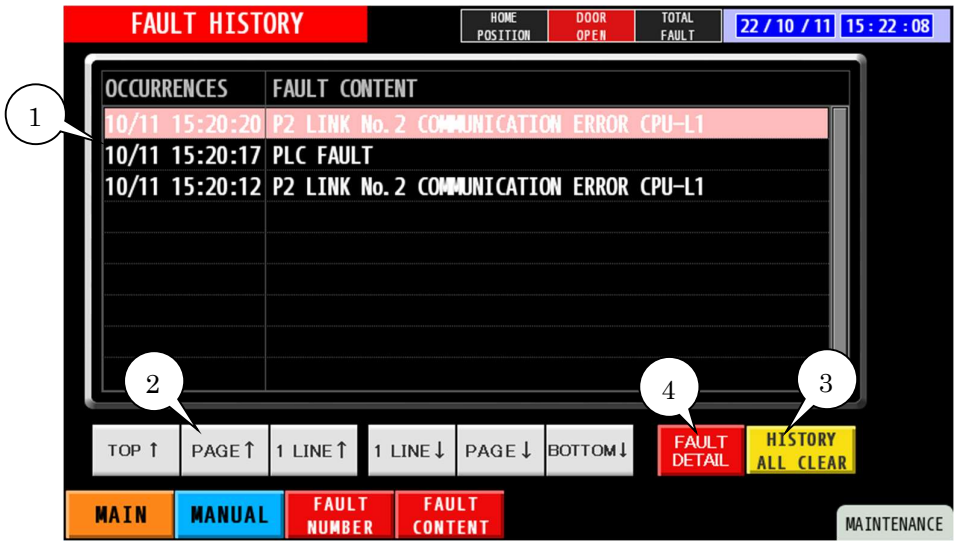
5-5-3. Fault Occurrence Counts



This screen displays the details of the error that occurred, the date and time, and the number of occurrences.

1) Displays the details of the error that occurred, the date and time, and the number of occurrences.
2) Cursor movement key
3) It is a key to delete all the abnormality history.

5-5-4. Fault History



This screen displays the details of the error that occurred and the date and time.

1) Displays the details of the error that occurred and the date and time.
2) Cursor movement key
3) It is a key to delete all the abnormality history.
4) Touch this switch to move to the FAULT DETAIL screen of the anomaly selected in 1).

5-5-5. ST1 Fault Description screen comment list

Comments on the Fault Description screen are allocated to internal relays M800 to M847 of PLC. When an internal relay turns ON, the comment corresponding to the internal relay is displayed.

If multiple internal relays should turn ON, all of them are displayed. In such a case, comments are displayed in the order of the relays having turned ON.

Abnormal contents comment table - 1

Device	Main comment	Detailed comment
M800	W/S rotation servo amplifier (VAR-1) fault.	Please turn the "Reset" button after confirms servo amplifier (VAR-1), and having removed a factor becoming abnorm.
M801	L/S rotation servo amplifier (VAR-3) fault.	Please turn the "Reset" button after confirms servo amplifier (VAR-3), and having removed a factor becoming abnorm.
M802	L/S servo converter (CONV-1) fault.	Please turn the "Reset" switch after confirms a converter (CONV-1), and having removed an abnormal factor.
M803	Light curtain was blocked.	Light curtain was blocked during machine actuating. Please turn the "RESET" switch.
M804	Air pressure fall.	Please turn a "RESET" switch after confirms air pressure and a pressure sensor (SP-030), and having confirmed a cause.
M805	L/S torque alarm	Please turn the "RESET" button after confirm work condition and torque amplifier power and torque sensor switch.
M806	Master start relay fault	Master start relay is abnormal. Please turn a “RESET” switch after confirms cause.
M807	Power supply relay fault	Power supply relay is abnormal. Please turn a “RESET” switch after confirms cause.
M808	Master start condition fault	Master start condition relay is abnormal. Please turn a “RESET” switch after confirms cause.
M809		
M810	W/S socket insertion confirm sensor fault.	Though a unit retrogresses, a confirmation sensor (LS-047) does ON. Please turn the "RESET" switch after having confirmed a cause.
M811		
M812	Lack of Tonner	A toner lacks. When toners decrease, please supply it. When toners do not decrease, please confirm a sensor (FLS-031).
M813	Auto run cycle time over.	Automatic driving was not finished. Please check the mechanical problems.

Abnormal contents comment table - 2

Device	Main comment	Detailed comment
M814	UNIT-10 CH1 communication time over	Confirm connection of connector of CH1 (UNIT-10)
M815	ST1 2D code reader data error	1) Check noise. 2) Check setting of 2D code reader.
M816	ST1 Traceability server time-out	1) Confirm traceability server. 2) Confirm LAN cables and switching hubs.
M817	UNIT-10 CH2 communication time over	Confirm connection of connector of CH2 (UNIT-10)
M818	QJ71C24N-R2 UNIT-10 Fault	QJ71C24N-R2 Abnormal signal turned ON from U-10
M819		
M820	Work Clamp forward and back time-out.	Work Clamp is taking too long to move forward or back. Please check if there are any mechanical problems such as damages or stuck on air piping. Check if there are any malfunction or position gap on detect sensors (LS-043,LS-044).
M821	W/S Socket Insert forward and back time-out.	W/S Socket Insert is taking too long to move forward or back. Please check if there are any mechanical problems such as damages or stuck on air piping. Check if there are any malfunction or position gap on detect sensors (LS-045,LS-046).
M822	W/S Rotation return to home position time-out.	Return the home position was not completed. Please turn the "RESET" switch after confirm servo amplifier (VAR-1) and the case.
M823	W/S Rotation positioning time-out.	W/S rotation positioning was not completed. Please turn the "RESET" switch after confirm servo amplifier (VAR-1) and the case.
M824	L/S position gap fault.	Please turn the "RESET" switch after confirm specification setup value and work condition.
M825		
M826	W/S Socket insert miss.	W/S socket insertion was not completed. Please turn the "RESET" switch after confirm machine and work.
M827	L/S insert miss	L/S socket insertion was not completed. Please turn the "RESET" switch after confirm machine and work.
M828	Work Clamp down and up time-out.	Work Clamp is taking too long to move down or up. Please check if there are any mechanical problems such as damages or stuck on air piping. Check if there are any malfunction or position gap on detect sensors (LS-042).

Abnormal contents comment table - 3

Device	Main comment	Detailed comment
M829	Unit Change right and left time-out.	Unit Change is taking too long to move right or left. Please check if there are any mechanical problems such as damages or stuck on air piping. Check if there are any malfunction or position gap on detect sensors (LS-049).
M830		
M831		
M832		
M833		
M834		
M835		
M836		
M837		
M838		
M839		
M840		
M841		
M842		
M843		
M844		
M845		
M846		
M847	PLC BATTERY FAULT	Please replace the PLC battery.

5-5-6. ST2 Fault Description screen comment list

Comments on the Fault Description screen are allocated to internal relays M1800 to M1847 of PLC. When an internal relay turns ON, the comment corresponding to the internal relay is displayed.

If multiple internal relays should turn ON, all of them are displayed. In such a case, comments are displayed in the order of the relays having turned ON.

Abnormal contents comment table - 1

Device	Main comment	Detailed comment
M1800	W/S rotation servo amplifier (VAR-3) fault.	Please turn the "Reset" button after confirms servo amplifier (VAR-3), and having removed a factor becoming abnorm.
M1801	L/S rotation servo amplifier (VAR-4) fault.	Please turn the "Reset" button after confirms servo amplifier (VAR-4), and having removed a factor becoming abnorm.
M1802	L/S servo converter (CONV-2) fault.	Please turn the "Reset" switch after confirms a converter (CONV-2), and having removed an abnormal factor.
M1803	Light curtain was blocked.	Light curtain was blocked during machine actuating. Please turn the "RESET" switch.
M1804	Air pressure fall.	Please turn a "RESET" switch after confirms air pressure and a pressure sensor (SP-025), and having confirmed a cause.
M1805	L/S torque alarm	Please turn the "RESET" button after confirm work condition and torque amplifier power and torque sensor switch.
M1806	Master start relay fault	Master start relay is abnormal. Please turn a “RESET” switch after confirms cause.
M1807	Power supply relay fault	Power supply relay is abnormal. Please turn a “RESET” switch after confirms cause.
M1808	Master start condition fault	Master start condition relay is abnormal. Please turn a “RESET” switch after confirms cause.
M1809	W/S Unit positioning motor overload	Thermal relay (FA-2) in control box tripped. Remove root cause and reset thermal relay and Reset button pushed.
M1810	W/S socket insertion confirm sensor fault.	Though a unit retrogresses, a confirmation sensor (LS-057) does ON. Please turn the "RESET" switch after having confirmed a cause.
M1811		
M1812	Lack of Tonner	A toner lacks. When toners decrease, please supply it. When toners do not decrease, please confirm a sensor (FLS-031).
M1813	Auto run cycle time over.	Automatic driving was not finished. Please check the mechanical problems.

Abnormal contents comment table - 2

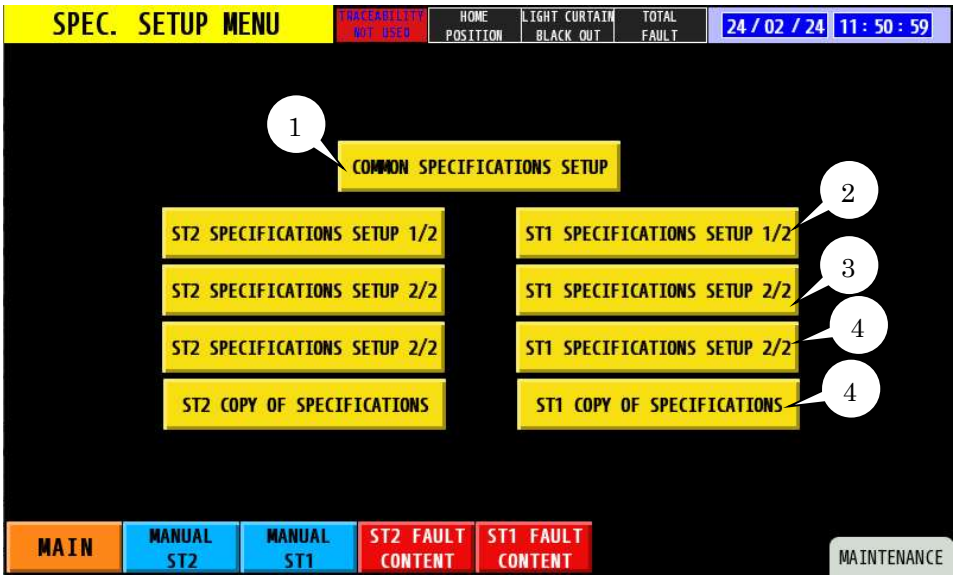
Device	Main comment	Detailed comment
M1814	UNIT-11 CH1 communication time over	Confirm connection of connector of CH1 (UNIT-11)
M1815	ST2 2D code reader data error	1) Check noise. 2) Check setting of 2D code reader.
M1816	ST2 Traceability server time-out	1) Confirm traceability server. 2) Confirm LAN cables and switching hubs.
M1817	UNIT-11 CH2 communication time over	Confirm connection of connector of CH2 (UNIT-11)
M1818	QJ71C24N-R2 UNIT-11 Fault	QJ71C24N-R2 Abnormal signal turned ON from U-11
M1819		
M1820	Work Clamp forward and back time-out.	Work Clamp is taking too long to move forward or back. Please check if there are any mechanical problems such as damages or stuck on air piping. Check if there are any malfunction or position gap on detect sensors (LS-053,LS-054).
M1821	W/S Socket Insert forward and back time-out.	W/S Socket Insert is taking too long to move forward or back. Please check if there are any mechanical problems such as damages or stuck on air piping. Check if there are any malfunction or position gap on detect sensors (LS-055,LS-056).
M1822	W/S Rotation return to home position time-out.	Return the home position was not completed. Please turn the "RESET" switch after confirm servo amplifier (VAR-3)and the case.
M1823	W/S Rotation positionig time-out.	W/S rotation positioning was not completed. Please turn the "RESET" switch after confirm servo amplifier (VAR-3) and the case.
M1824	L/S position gap fault.	Please turn the "RESET" switch after confirm specification setup value and work condition.
M1825		
M1826	W/S socket insert miss.	W/S socket insertion was not completed. Please turn the "RESET" switch after confirm machine and work.
M1827	L/S socket insert miss.	L/S socket insertion was not completed. Please turn the "RESET" switch after confirm machine and work.
M1828	Work Clamp down and up time-out.	Work Clamp is taking too long to move down or up. Please check if there are any mechanical problems such as damages or stuck on air piping. Check if there are any malfunction or position gap on detect sensors (LS-052).

Abnormal contents comment table - 3

Device	Main comment	Detailed comment
M1829	Unit Change right and left time-out.	Unit Change is taking too long to move right or left. Please check if there are any mechanical problems such as damages or stuck on air piping. Check if there are any malfunction or position gap on detect sensors (LS-059).
M1830		
M1831		
M1832		
M1833		
M1834		
M1835		
M1836		
M1837		
M1838		
M1839		
M1840		
M1841		
M1842		
M1843		
M1844		
M1845		
M1846		
M1847		

5-6. Specifications Setting Menu

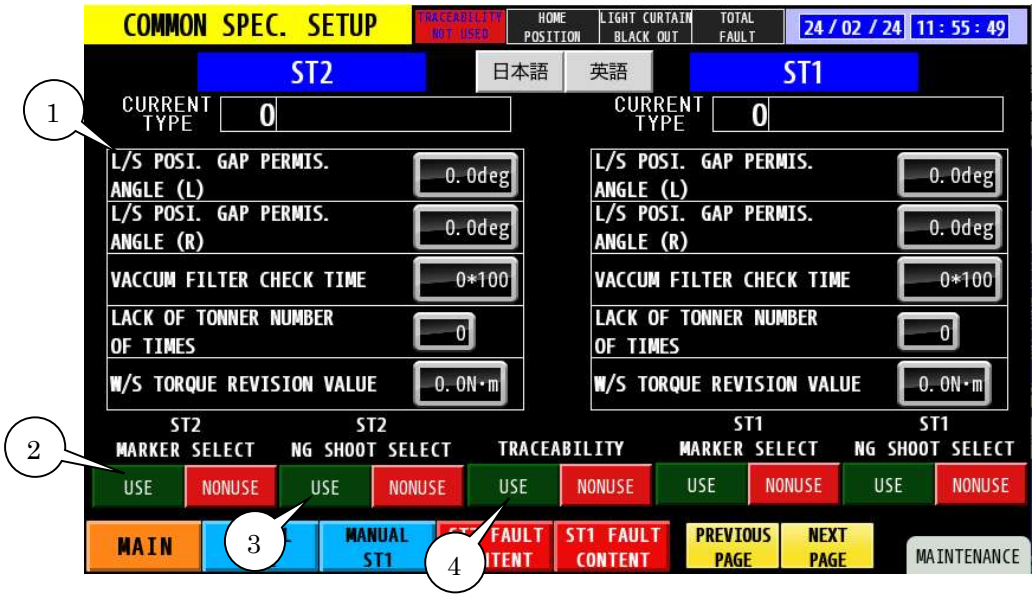
The password is necessary to shift to this standard setting menu screen.



1) Common Specifications Set up
   Changing to Later description 5-7 common specifications setting window
2) Specifications Setup Screen 1/3
   Changing to Later description 5-8 specifications set up screen 1/3
3) Specifications Setup Screen 2/3
   Changing to Later description 5-9 specifications set up screen 2/3
4) Specifications Setup Screen 3/3
   Changing to Later description 5-10 specifications set up screen 3/3
5) Copy of Specifications
   Changing to later description 5-11 copy screen

5-7. Common Specifications Setup Screen

With this screen, set setting items memorizing in PLC.
Each setting is registered with every work type.
Attention) Change of a setting value cannot be performed during automatic operation.



1) Expression column of various set points

1)-1. L/S posi.gap permis angle
Set a permission angle of a position as of a thing of W/S and L/S.

1)-2. L/S servo oil supply time
Set the oil supply time of a load shaft servomotor.
Start *100 times input values auto driving.
It is the oil supply time and gets impossible to start.

1)-3. Vacuum filter check time
Set the time checking an air blow filter.
Start *100 times input values auto driving.
It is the filter check time and gets impossible to start.

1)-4. Lack of tonner number of times
Apply this setting times after a quantity of oil confirmation sensor did OFF.
Become lack of toner abnormality, and get impossible to start auto driving.

1)-5. W/S torque revision value
Set revision value of W/S torque.

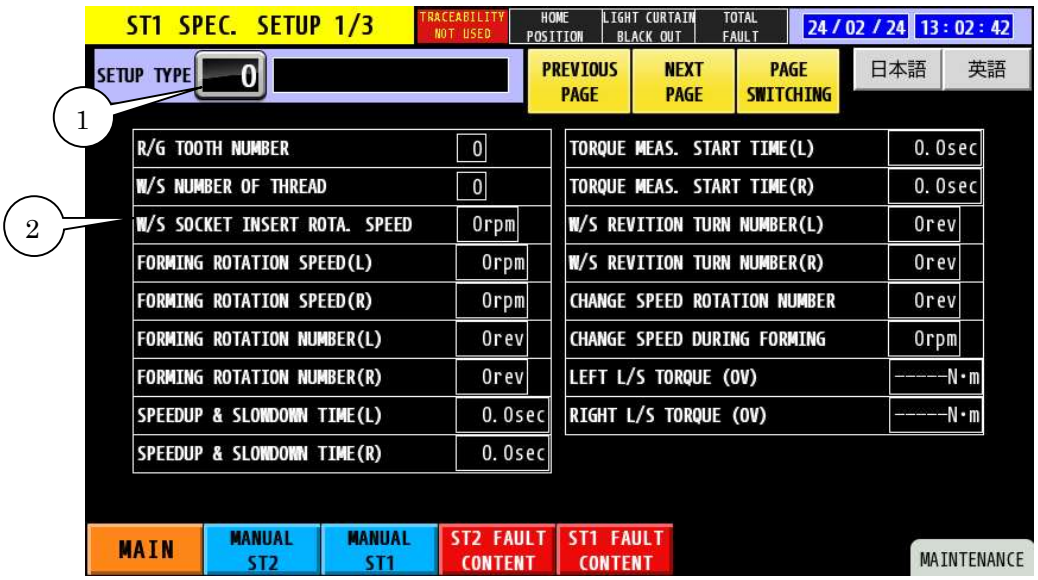
2) Marker check Use/No use switch

3) NG shoot Use/No use switch

4) Traceability Use/No use switch

5-8. Specifications setup screen 1/3

With this screen, set setting items memorizing in PLC.
Each setting is registered with every work type.
As for the setting of ST2, content becomes similar.
Attention) Change of a setting value cannot be performed during automatic operation.



1) The work type number and the name selected for monitor currently is displayed.

2) Expression column of various set points

2)-1. R/G tooth number
Set the number of the teeth of a reduction gear.
Decide rotation speed of a W/S shaft. Please set it with care.

2)-2. W/S number of start
Set the number of the screws of a W/S shaft.
Decide rotation speed of a W/S shaft. Please set it with care.

2)-3. W/S socket insert rota. speed
Set rotation speed at the time of the W/S serration socket insertion.

2)-4. Forming rotation speed (L)
Set rotation speed of the left forming. This speed becomes the speed that R/G turns.

2)-5. Forming rotation speed (R)
Set rotation speed of the right forming. This speed becomes the speed that R/G turns.

2)-6. Forming rotation number (L)
Set the number of the rotations of the left forming. This speed becomes the number of the rotations of R/G.

2)-7. Forming rotation number (R)
Set the number of the rotations of the right forming. This speed becomes the number of the rotations of R/G.

2)-8. Speedup & slowdown time (L)
Set rotation picking up speed and slowing down time of a W/S shaft.

2)-9. Speedup & slowdown time (R)
Set rotation picking up speed and slowing down time of a W/S shaft.

2)-10. Torque meas. start time(L)
Set start time to judge the left forming torque to be it. This setting passes in picking up speed and slowing down time after W/S began to turn around and empties.

2)-11. Torque meas. start time(R)
Set start time to judge the right forming torque to be it. This setting passes in picking up speed and slowing down time after W/S began to turn around and empties.

2)-12. W/S Revision turn number(L)
Set a torque revision rotation number during the left forming. Revise torque every this setting.

2)-13. W/S Revision turn number(R)
Set a torque revision rotation number during the right forming. Revise torque every this setting.

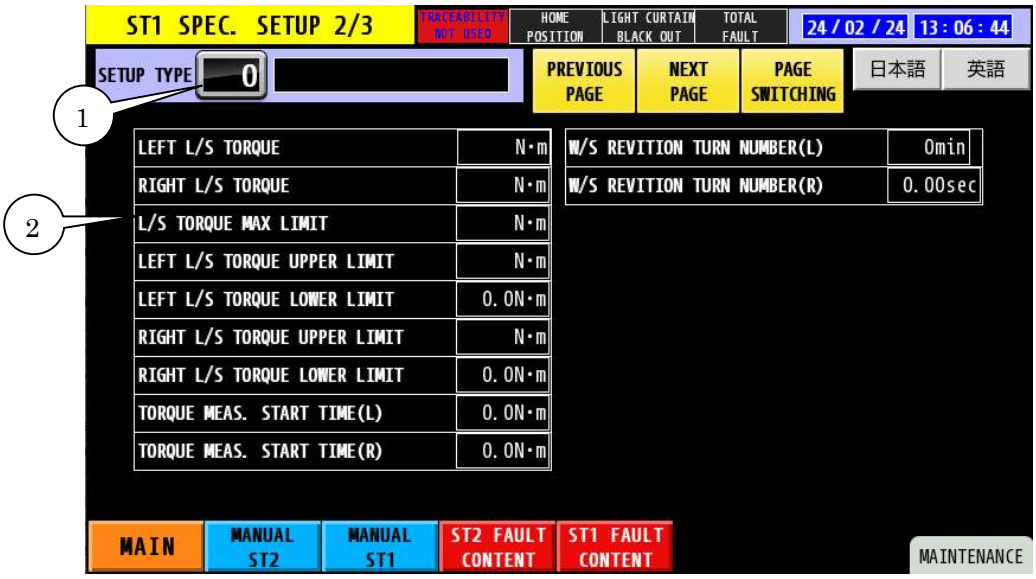
2)-14. Change speed rotation number
Set the number of the turns to play by speed changing during forming. This setting becomes the number of the turns of L/S

2)-15. Change speed during Forming
Set speed to make speed changing during forming. This setting becomes turn speed of L/S.

2)-16. Left L/S load torque(0V)
Set 0V torque at the time of the left rotation. it every automatic driving auto measure, and change setting.

2)-17. Right L/S load torque(0V)
Set 0V torque at the time of the right rotation. it every automatic driving auto measure, and change setting.

5-9. Specifications setup screen 2/3



1) The work type number and the name selected for monitor currently is displayed.

2) Expression column of various set points

2)-1. Left L/S torque

Set target torque value at the time of the left forming.

2)-2. Right L/S torque

Set target torque value at the time of the right forming.

2)-3. L/S load torque max limit

Set the upper limit of load torque.

Become abnormal when beyond this set point.

2)-4. Left L/S torque upper limit

Set the load torque upper limit during the left forming.

When load torque is beyond this set point after torque judgment start time, become load torque NG.

2)-5. Left L/S torque lower limit

Set the load torque lower limit during the left forming. When load torque is less than this set point after torque judgment start time, become load torque NG.

2)-6. Right L/S torque upper limit

Set the load torque upper limit during the right forming. When load torque is beyond this set point after torque judgment start time, become load torque NG.

2)-7. Right L/S torque lower limit

Set the load torque lower limit during the right forming. When load torque is less than this set point after torque judgment start time, become load torque NG.

2)-8. Left L/S average torque upper limit

Set the load average torque upper limit during the left forming. When load torque is beyond this set point after torque judgment start time, become load torque NG.

2)-9. Left L/S average torque lower limit
Set the load average torque lower limit during the left forming. When load torque is less than this set point after torque judgment start time, become load torque NG.

2)-10. Right L/S average torque upper limit
Set the load average torque upper limit during the right forming. When load torque is beyond this set point after torque judgment start time, become load torque NG.

2)-11. Right L/S average torque lower limit
Set the load average torque lower limit during the right forming. When load torque is less than this set point after torque judgment start time, become load torque NG.

2)-12. W/S torque lower limit
Set the input shaft torque lower limit during forming. When input shaft torque is less than this setting after torque judgment start time, become input shaft torque NG.

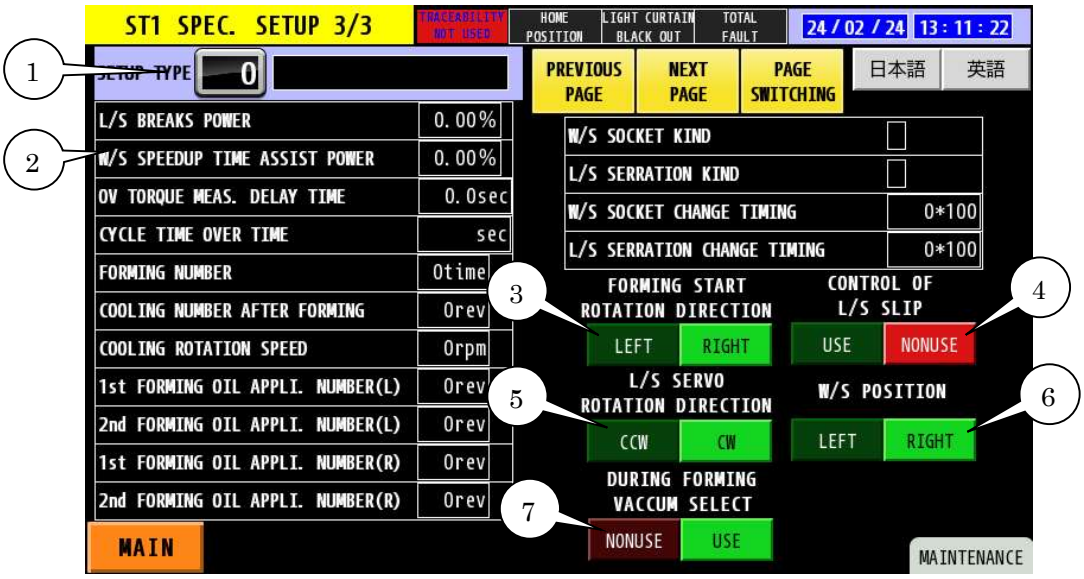
2)-13. W/S torque upper limit
Set the input shaft torque upper limit during forming. When input shaft torque is beyond this setting after torque judgment start time, become input shaft torque NG.

2)-14. L/S climate cycle interval
Set distance to perform genial climate driving of L/S rotation sorvo. When does not start auto driving in current setting time, get impossible to machine a work unless starts genial climate driving.

2)-15. L/S climate cycle time
Set rotation time at the time of genial climate driving.

5-10. Specifications setup screen 3/3

With this screen, set setting items memorizing in PLC.
Each setting is registered with every work type.
As for the setting of ST2, content becomes similar.
Attention) Change of a setting value cannot be performed during automatic operation.



1) The work type number and the name selected for monitor currently is displayed.

2) Expression column of various set points

2)-1. L/S breaks power
Set power to brake so that L/S does not do attendants at the time of the W/S socket insertion.

2)-2. W/S speedup time assist power
Set power to assist for at the time of W/S rotation acceleration so that load torque does not become high.

2)-3. 0V torque meas. delay time
Measure 0V torque after W/S rotation acceleration completion, but set the measurement start delay line.

2)-4. Cycle time over time
Set a cycle time over of auto driving. If cycle time becomes longer than this setting, become abnormal.

2)-5. Forming number
Set forming coming and going times. Be one coming and going by right and left forming rotation completion.

2)-6. Cooling number after forming
Set a cooling rotation number after forming. This sets the number of the rotations of L/S.

2)-7. Cooling rotation speed
Set cooling rotation speed after forming. This becomes the speed that L/S turns.

2)-8. 1st Forming oil appli. number (L)
Set the number of the turns to apply a tonner in the left forming first half.

2)-9. 2nd Forming oil appli. number (L)
Set the number of the turns to apply a toner in the left familiarity latter half.

2)-10. 1st Forming oil appli. number (R)
Set the number of the turns to apply a tonner in the right forming first half.

2)-11. 2nd Forming oil appli. number (R)
Set the number of the turns to apply a toner in the Right familiarity latter half.

2)-12. W/S socket kind
Set a kind of a W/S socket with a number. Memorize W/S use times every set kind here.

2)-13. L/S serration kind
Set a kind of L/S Serration. Memorize use times of L/S Serration every set kind here.

2)-14. W/S socket change timing
Set the exchange time of a W/S socket. It is the exchange time when starts this *100 times setting auto driving.

2)-15. L/S serration change timing
Set the exchange time of a L/S serration. It is the exchange time when starts this *100 times setting auto driving.

3) The selection that is given priority to right and left at forming start time
Set forming rotatory direction.

4) L/S position gap detection selection
Perform gap detection of a position with a W/S shaft and a L/S shaft when chooses ON. Do not detect a position gap when turns it off.

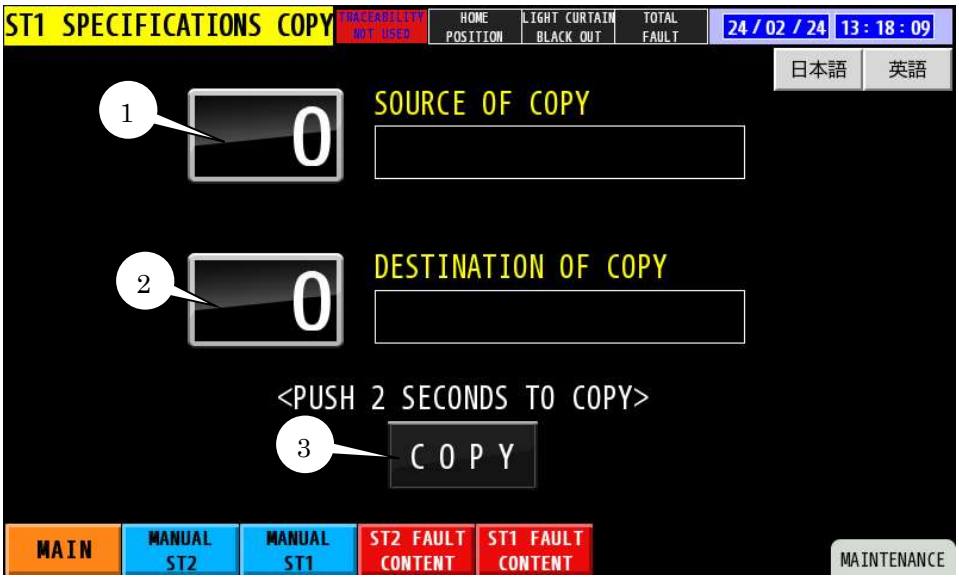
5) L/S load torque rotatory direction selection
Choose rotatory direction of load torque. Assist when changes rotatory direction.

6) W/S position selection
Seeing from the system front, set a position of W/S.

7) During Forming Vaccum selection

5-11. Copy of Specifications

Changing to this screen is possible only from 5-8.Specifications Setup Screen.
With this screen, copying settings of arbitrary work type into another arbitrary work type is possible. But the name of the work type is not copied.
For adding a new work type, it is efficient to change only diversity entries with 5-8.Specifications Setup Screen after having used this function. If it is a masterwork, it is efficient all the more.



1) "COPY SOURCE SELECT"
The type name of selected source of copy is displayed.

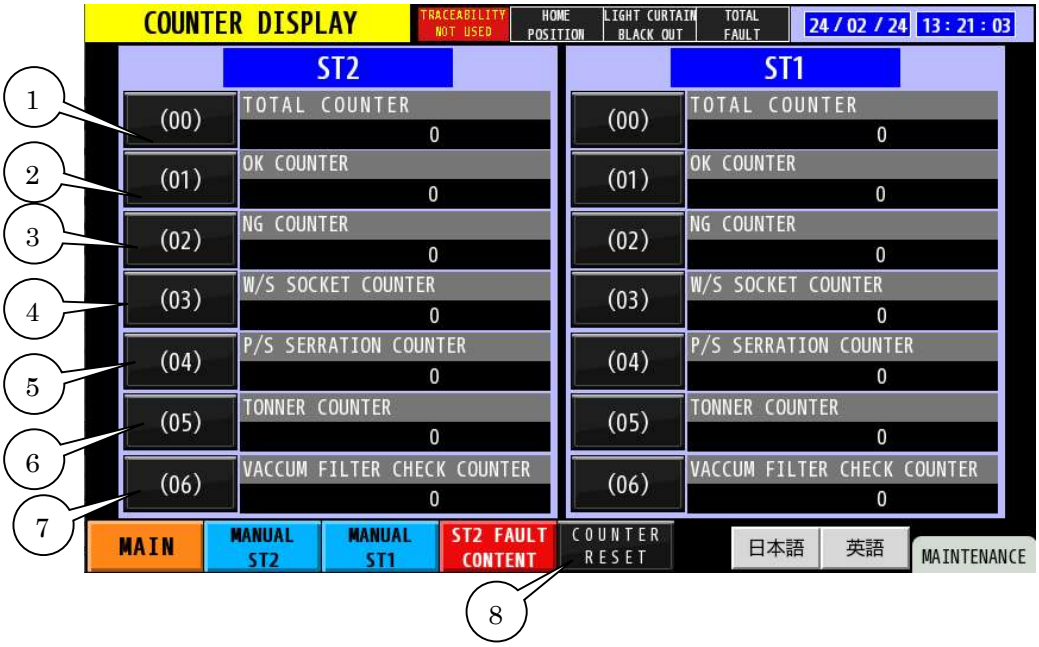
2) "DESTINATION OF COPY"
The type name of selected destination of copy is displayed.

3) Conduct a copy.
Copy is conducted by selecting the source of copy and destination of copy, then press "COPY" button for 2 seconds.

Be careful not to accidentally copy to a different type number.

5-12. Counter screen

Display various counters with this screen.



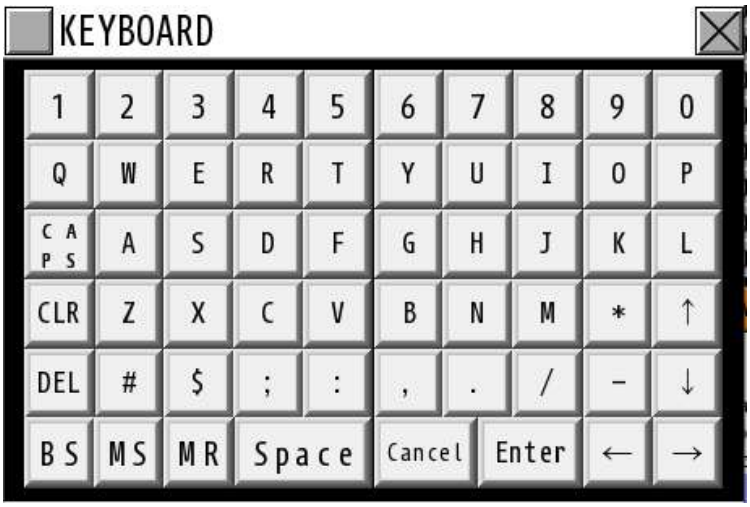
1) Total Counter
   Display times to Total of operations completion.
2) OK Counter
   Display times to OK of operations completion.
3) NG Counter
   Display times to NG of operations completion.
4) W/S Socket Counter
   Display use times of a W/S socket.
5) Serration Counter
   Display use times of a Serration.
6) Toner application Counter
   OFF is condition that did, and quantity of toner confirmation sensor(FLS220) displays the times that applied.
7) Vacuum Filter Counter
   Display use times of a Vacuum.
8) Counter Reset PB
   After selecting the number to reset, press the reset button for 2 seconds to reset it to 0.

5-13. Type Name Registration

Register a name of each work type (0 to 99) with this screen.
Arbitrary characters in a keypad window are used to register a name of each work type.
(Maximum 16 characters)

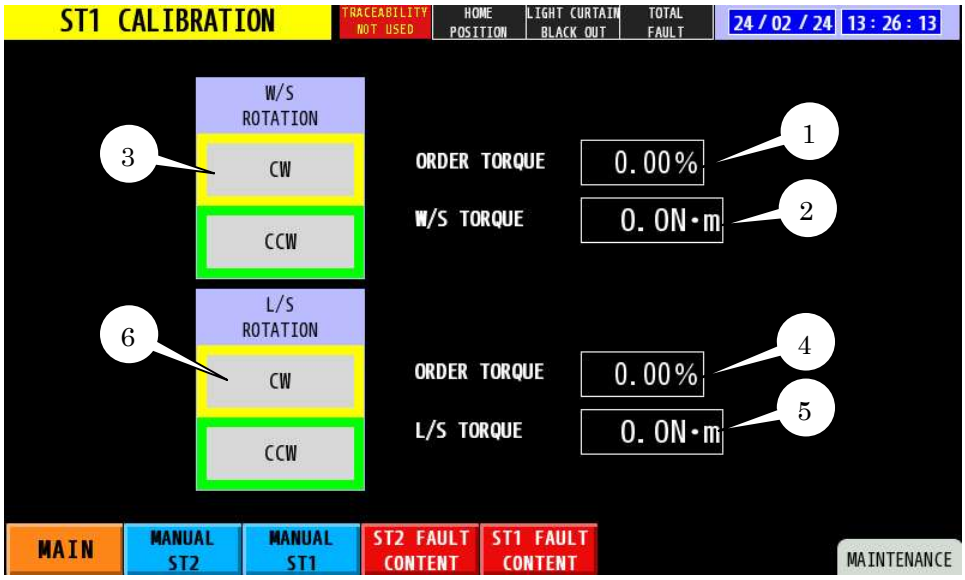


1) Each work type name indication column
Touch the each work type name indication column, and below registered names and a cursor are displayed.



5-14. Load Torque Calibration Screen

Display a measurement result of load torque with this screen.



1) W/S Rotation order Torque
I set torque value to operate a W/S rotation at a percentage.

2) W/S Torque
It display torque value depending on a servomotor

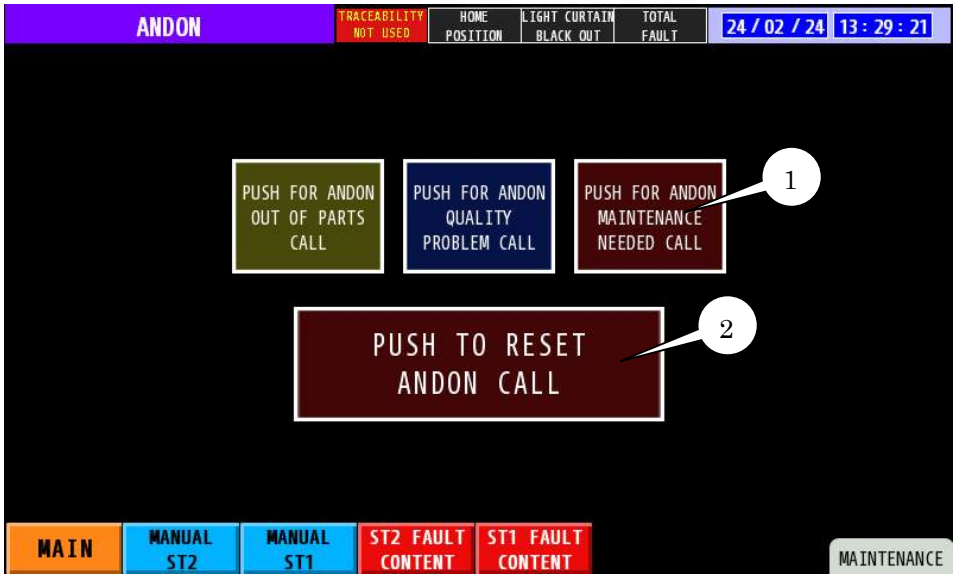
3) W/S Rotation Manual Operation Selection Switch
Can choose a selection select switch at the time of manual operation for a cycle. Can work with a "manual operation" illuminated push button switch and a "Manual Operation Home Position side" illuminated push button switch after selection with this switch.

4) L/S Rotation order Torque
I set torque value to operate a L/S rotation at a percentage.

5) L/S Torque
It display torque value depending on a torque meter.

6) L/S Rotation Manual Operation Selection Switch
Can choose a selection select switch at the time of manual operation for a cycle. Can work with a "manual operation" illuminated push button switch and a "Manual Operation Home Position side" illuminated push button switch after selection with this switch.

5-15. ANDON Screen



1) Various call buttons
   This is a call button for material shortages, quality, and maintenance.
2) ANDON call reset button
   Resets the call button that was turned on in step 1).

6. Calibration method of strain amplifier

Calibration method of strain amplifier as follows.

1) Turn on the power switch of the machine.
2) Put power switch of strain amplifier. Torque sensor turns sensor body switch into the on side.
3) Load (force) is not added to torque sensor and load cell sensor.
4) Filter regulating apparatus selects 〔 10Hz 〕.
5) Push a button of Bal once.
6) Confirm rated value mentioned in test record of result of sensor and value of rated output. Please set rated output (mV/V) value with sum rotary switch setting calibration (CAL) value.
7) Turn screen of touch panel into SENSOR MONITOR. Do control of afterward while looking at input data of each sensor with screen of SENSOR MONITOR.
8) Turn toggle switch of CAL polarity into 〔 OFF 〕 (the center). Adjust it with BAL button so that indicated value of touch panel becomes zero.
9) Turn toggle switch of CAL polarity into 〔 + 〕 side. Adjust gain with volume of FINE GAIN so that indicated value of touch panel becomes one-half of rated capacity (unit, N･m).
10) Several times repetition identifies 8) as 9).
11) If calibration was over, turn toggle switch of CAL polarity into 〔 OFF 〕 (the center).

Torque sensor

	ST1	ST2
Rated capacity (N)	150 N･m	150 N･m
Rated output (mV/V)	1.502 mV/V	1.504 mV/V
Calibration set point	1502	1504

Display of touch panel

	ST1	ST2
CAL polarity [+]	100.0 N･m	100.0 N･m
CAL polarity [OFF]	0.0 N･m	0.0 N･m
CAL polarity [−]	100.0 N･m	100.0 N･m